

CCNA 200-301 Complete Lab Manual

100 Hands-On Labs for Packet Tracer & GNS3

From Beginner to Advanced

Created for CCNA Candidates
Based on Official Exam Blueprint

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Introduction

How to Use This Manual

This manual contains **100 structured labs** designed to prepare you for the CCNA 200-301 exam. Labs progress from basic configuration to complex troubleshooting scenarios.

Prerequisites

- Cisco Packet Tracer (v8.0+) or GNS3 with Cisco IOS images
- Basic understanding of networking concepts
- 2-3 months of dedicated practice (1-2 hours daily)

Lab Structure

Each lab includes:

- **Objective:** What you will accomplish
- **Topology:** Device layout (text-based diagram)
- **Configuration Steps:** Commands with explanations
- **Verification:** Show commands and expected output
- **Troubleshooting:** Common issues and solutions

Conventions

- `monospaced text` = Commands and output
- **Bold text** = Important concepts
- *Italic text* = Variables (replace with your values)

1 Beginner Level: Foundation Labs

Week 1-2: Basic Navigation and Configuration

1.1 Lab 1: Basic Network Topology Setup

Objective

Create a simple network with 2 PCs and 1 switch, then connect them properly.

Topology

```
[PC1] --- [Switch1] --- [PC2]
(Fa0)      (Fa0/1)      (Fa0/2)
```

Configuration Steps

1. Open Packet Tracer and add:
 - 1 x 2960 Switch
 - 2 x Generic PCs
2. Connect devices using Copper Straight-Through cables:
 - PC1 Fa0 → Switch Fa0/1
 - PC2 Fa0 → Switch Fa0/2
3. Verify link lights turn green (indicates Layer 1 connectivity)

Verification

```
1 ! On switch:
2 Switch> show interfaces status
3 Port      Name      Status      Vlan      Duplex  Speed  Type
4 Fa0/1      Name      connected   1         auto    auto   10/100BaseTX
5 Fa0/2      Name      connected   1         auto    auto   10/100BaseTX
```

Troubleshooting

- If lights are amber/orange → Check cable type (use straight-through for PC to Switch)
- If lights are off → Verify both devices are powered on

1.2 Lab 2: Connecting PCs via Switch

Objective

Establish Layer 2 connectivity between two PCs through a single switch.

Topology

Same as Lab 1.

Configuration Steps

1. Configure IP addresses on PCs:
 - PC1: 192.168.1.1/24
 - PC2: 192.168.1.2/24
2. No switch configuration needed initially (default VLAN 1)

Verification

```
1 ! On PC1:
2 C:\> ping 192.168.1.2
3 Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
4 Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

Expected Result

Ping success rate: 100%

1.3 Lab 3: Basic Router Configuration

Objective

Configure a router's hostname, interfaces, and verify connectivity.

Topology

```
[PC1] --- [Router1] --- [PC2]
.1       .254       .254       .2
(192.168.1.0/24)   (192.168.2.0/24)
```

Configuration Steps

```
1 Router> enable
2 Router# configure terminal
3 Router(config)# hostname R1
4 R1(config)# interface gigabitEthernet 0/0
5 R1(config-if)# ip address 192.168.1.254 255.255.255.0
6 R1(config-if)# no shutdown
7 R1(config-if)# exit
8 R1(config)# interface gigabitEthernet 0/1
9 R1(config-if)# ip address 192.168.2.254 255.255.255.0
10 R1(config-if)# no shutdown
11 R1(config-if)# end
12 R1# write memory
```

Verification

```
1 R1# show ip interface brief
2 Interface                IP-Address      OK? Method Status Protocol
3 GigabitEthernet0/0       192.168.1.254   YES manual up      up
4 GigabitEthernet0/1       192.168.2.254   YES manual up      up
```

1.4 Lab 4: Assigning IP Addresses to PCs

Objective

Learn different methods of assigning IP addresses to end devices.

Configuration Steps

1. Static IP Assignment (Windows CLI):

```
1 PC> ipconfig 192.168.1.10 255.255.255.0 192.168.1.1
2
```

2. Static IP via GUI:

- Desktop → IP Configuration
- Enter: IP Address, Subnet Mask, Default Gateway

3. DHCP Assignment (after Lab 55):

```
1 PC> ipconfig /dhcp
2
```

Verification

```
1 PC> ipconfig /all
2 FastEthernet0 Connection:
3     IP Address: 192.168.1.10
4     Subnet Mask: 255.255.255.0
5     Default Gateway: 192.168.1.1
6     DHCP Enabled: No
```


1.5 Lab 5: Configuring Hostnames on Devices

Objective

Properly name network devices for identification and management.

Configuration Steps

```
1 Switch> enable
2 Switch# configure terminal
3 Switch(config)# hostname SW1
4 SW1(config)# exit
5 SW1#
6
7 Router> enable
8 Router# configure terminal
9 Router(config)# hostname R1
10 R1(config)# exit
11 R1#
```

Naming Conventions

- **Switches:** SW-[Location]-[Number] (e.g., SW-MDF-01)
- **Routers:** R-[Location]-[Number] (e.g., R-EDGE-01)
- **PCs:** PC-[VLAN]-[Number] (e.g., PC-10-05)

Verification

```
1 SW1# show running-config | include hostname
2 hostname SW1
```

1.6 Lab 6: Setting Up Console Access

Objective

Secure and configure console port access for out-of-band management.

Configuration Steps

```
1 SW1> enable
2 SW1# configure terminal
3 SW1(config)# line console 0
4 SW1(config-line)# password cisco123
5 SW1(config-line)# login
6 SW1(config-line)# exec-timeout 5 30
7 SW1(config-line)# logging synchronous
8 SW1(config-line)# end
9 SW1# write memory
```

Configuration Explanations

- password cisco123 - Sets console password
- login - Enables password authentication
- exec-timeout 5 30 - Times out after 5 minutes, 30 seconds
- logging synchronous - Prevents console messages from interrupting typing

Verification

```
1 ! Exit and reconnect to console
2 SW1# exit
3
4 Press Enter
5 Password: cisco123
6 SW1>
```

1.7 Lab 7: Configuring Enable Password & Secret

Objective

Understand the difference between enable password (unencrypted) and enable secret (encrypted).

Configuration Steps

```
1 R1> enable
2 R1# configure terminal
3 R1(config)# enable password cisco
4 R1(config)# enable secret cisco123
5 R1(config)# exit
6 R1# exit
7
8 ! Re-enter privileged mode
9 R1> enable
10 Password: cisco123
11 R1#
```

Verification - Viewing the Difference

```
1 R1# show running-config | include enable
2 enable secret 5 $1$Q8I5$LQYqZxKdV8kKo7HkPQkP10
3 enable password cisco
```

Key Points

- **enable secret** uses MD5 hashing (Type 5)
- **enable password** is stored in plain text (Type 0)
- If both configured, enable secret takes precedence
- Always use enable secret

1.8 Lab 8: Saving Configuration (Startup vs Running)

Objective

Understand the difference between running-config (RAM) and startup-config (NVRAM).

Key Concepts

- **Running-config:** Active configuration in RAM (lost on reboot)
- **Startup-config:** Saved configuration in NVRAM (loaded on boot)

Configuration Steps

```
1 ! View both configurations
2 R1# show running-config
3 R1# show startup-config
4
5 ! Save running to startup
6 R1# copy running-config startup-config
7 Destination filename [startup-config]? [Enter]
8 Building configuration...
9 [OK]
10
11 ! Alternative methods
12 R1# write memory
13 R1# copy run start
14
15 ! Erase startup config (for lab cleanup)
16 R1# erase startup-config
17 Erasing the nvram filesystem will remove all configuration files!
   Continue? [confirm]
18 [OK]
```

Verification

```
1 R1# show version | include image
2 System image file is "flash:c2900-universalk9-mz.SPA.157-3.M3.bin"
3 Configuration register is 0x2102
```

1.9 Lab 9: Static IP Configuration on PCs

Objective

Manually configure IP settings on end devices without DHCP.

Topology

```
[PC1] --- [Switch] --- [PC2]
.10/24           .20/24
```

Configuration Steps

1. On PC1 (Windows Command Line):

```
1 PC> ipconfig 192.168.1.10 255.255.255.0 192.168.1.1
2
```

2. On PC2:

```
1 PC> ipconfig 192.168.1.20 255.255.255.0 192.168.1.1
2
```

3. Verify DNS setting (optional):

```
1 PC> ipconfig /dns 8.8.8.8
2
```

Verification

```
1 PC1> ping 192.168.1.20
2 Pinging 192.168.1.20 with 32 bytes of data:
3 Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
4 Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
5
6 PC1> arp -a
7 Interface: 192.168.1.10 --- 0x3
8   Internet Address      Physical Address      Type
9   192.168.1.20          aaaa.bbbb.cccc       dynamic
```

1.10 Lab 10: Basic Ping Connectivity Test

Objective

Use ping to verify Layer 3 connectivity and troubleshoot issues.

Ping Command Options

```
1 ! Basic ping
2 PC> ping 192.168.1.1
3
4 ! Extended ping (on router)
5 R1# ping
6 Protocol [ip]:
7 Target IP address: 192.168.2.1
8 Repeat count [5]: 10
9 Datagram size [100]: 1500
10 Timeout in seconds [2]:
11 Extended commands [n]: y
12 Source address or interface: 192.168.1.254
13 Type of service [0]:
14 Set DF bit in IP header? [no]:
15 Validate reply data? [no]:
16 Data pattern [0xABCD]:
17 Loose, Strict, Record, Timestamp, Verbose[none]:
18 Sweep range of sizes [n]:
```

Interpreting Results

!!!! Success (0% loss)
..... No response (100% loss)
!..! Intermittent loss (congestion)
U.U.U Destination unreachable (routing issue)
Q.Q.Q Source quench (congestion)

Troubleshooting Matrix

Symptom	Probable Cause
"Request timed out"	Wrong IP, switch port down, ACL blocking
"Destination unreachable"	Default gateway missing, wrong subnet mask
"TTL expired"	Routing loop, TTL too low
"Reply from 192.168.1.254: Destination host unreachable"	No ARP entry, wrong VLAN

1.11 Lab 11: Using 'show' Commands Basics

Objective

Master essential show commands for monitoring and troubleshooting.

Essential Show Commands

```
1 ! Interface status
2 R1# show ip interface brief
3 R1# show interfaces
4 R1# show interfaces description
5
6 ! Routing
7 R1# show ip route
8 R1# show ip route [network]
9 R1# show ip protocols
10
11 ! System information
12 R1# show version
13 R1# show running-config
14 R1# show startup-config
15 R1# show clock
16
17 ! Layer 2
18 SW1# show mac address-table
19 SW1# show vlan brief
20 SW1# show spanning-tree
21
22 ! Security
23 R1# show ip ssh
24 R1# show users
25 R1# show privilege
26
27 ! Filtering output
28 R1# show running-config | include hostname
29 R1# show running-config | section line
30 R1# show running-config | begin interface
31 R1# show ip interface brief | exclude unassigned
```

Output Filtering Examples

```
1 ! Show only FastEthernet interfaces
2 R1# show ip interface brief | include FastEthernet
3
4 ! Show lines 5-10
5 R1# show running-config | include FastEthernet
6
7 ! Count occurrences
8 R1# show running-config | count interface
```

1.12 Lab 12: Switch MAC Address Table Exploration

Objective

Understand how switches learn MAC addresses and forward frames.

Topology

```

[PC1] --- [SW1] --- [PC2]
MAC A      MAC B

```

Configuration Steps

1. Connect PC1 and PC2 to SW1
2. Configure IP addresses on PCs
3. Generate traffic with ping

Verification

```

1 ! View empty MAC table (before traffic)
2 SW1# show mac address-table
3      Mac Address Table
4 -----
5 Vlan      Mac Address      Type      Ports
6 ----      -
7
8 ! Generate traffic from PC1 to PC2
9 PC1> ping 192.168.1.2
10
11 ! View learned MAC addresses
12 SW1# show mac address-table
13      Mac Address Table
14 -----
15 Vlan      Mac Address      Type      Ports
16 ----      -
17 1         aaaa.bbbb.cccc   DYNAMIC   Fa0/1
18 1         dddd.eeee.ffff   DYNAMIC   Fa0/2
19
20 ! Clear MAC table
21 SW1# clear mac address-table dynamic
22
23 ! Set static MAC entry
24 SW1# configure terminal
25 SW1(config)# mac address-table static aaaa.bbbb.cccc vlan 1 interface
    fa0/1

```

MAC Learning Process

1. PC1 sends frame to PC2 (dest MAC = PC2's MAC)
2. Switch records PC1's MAC and port (Fa0/1)
3. Switch floods frame out all ports (except Fa0/1)

4. PC2 responds, switch records PC2's MAC (Fa0/2)
5. Subsequent frames are forwarded directly (no flooding)

1.13 Lab 13: Configuring Duplex and Speed

Objective

Manually set interface speed and duplex to resolve negotiation issues.

Configuration Steps

```
1 SW1> enable
2 SW1# configure terminal
3 SW1(config)# interface fastEthernet 0/1
4 SW1(config-if)# speed 100
5 SW1(config-if)# duplex full
6 SW1(config-if)# no shutdown
7
8 ! Configure matching settings on connected device
9 ! On PC1 or connected router, set same speed/duplex
```

Speed/Duplex Options

Speed	Values	Best Practice
10	10 Mbps	Legacy only
100	100 Mbps	FastEthernet
1000	1 Gbps	GigabitEthernet
auto	Auto-negotiate	Default (recommended)

Duplex Options

Type	Use Case	Description
Full-duplex	Switch to switch	Send/receive simultaneously
Half-duplex	Hub or old devices	One direction at a time
auto	Most modern links	Negotiate best mode

Troubleshooting

```
1 ! Check for duplex mismatch
2 SW1# show interfaces fa0/1 | include duplex
3 Full-duplex, 100Mb/s, media type is 10/100BaseTX
4
5 ! Look for collisions (half-duplex only)
6 SW1# show interfaces fa0/1 | include collisions
7 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
```

Verification

```
1 SW1# show interfaces status
2 Port      Name      Status      Vlan      Duplex  Speed  Type
3 Fa0/1      connected 1           full      100     10/100BaseTX
```

1.14 Lab 14: Introduction to CDP (Cisco Discovery Protocol)

Objective

Use CDP to discover neighboring Cisco devices automatically.

Topology

```
[R1] --- [SW1] --- [R2]
(g0/0)   (fa0/1)   (g0/1)
```

Configuration Steps

```
1 ! CDP is enabled by default on Cisco devices
2 ! View CDP neighbors
3 R1# show cdp neighbors
4 Capability Codes: R - Router, S - Switch, H - Host
5 Device ID      Local Intrfce   Holdtme    Capability  Platform
6 SW1            Gig 0/0         156        S I         WS-C2960   Fa0
7 /1
8 ! Detailed neighbor information
9 R1# show cdp neighbors detail
10
11 ! View CDP traffic statistics
12 R1# show cdp traffic
13
14 ! CDP timers (default: 60 sec hello, 180 sec hold)
15 R1# show cdp interface
```

CDP Configuration

```
1 ! Disable CDP globally
2 R1(config)# no cdp run
3
4 ! Re-enable CDP
5 R1(config)# cdp run
6
7 ! Disable CDP on specific interface
8 R1(config)# interface gigabitEthernet 0/0
9 R1(config-if)# no cdp enable
10
11 ! Change CDP timers
12 R1(config)# cdp timer 30
13 R1(config)# cdp holdtime 120
```

Verification

```
1 R1# show cdp neighbors
2 R1# show cdp entry *
```

1.15 Lab 15: Basic Switch Security (Port Shutdown)

Objective

Secure unused switch ports by administratively shutting them down.

Configuration Steps

```
1 SW1> enable
2 SW1# configure terminal
3
4 ! Shutdown multiple unused ports
5 SW1(config)# interface range fastEthernet 0/10-24
6 SW1(config-if-range)# shutdown
7 SW1(config-if-range)# description ***UNUSED PORT - ADMIN DOWN***
8
9 ! Verify
10 SW1(config-if-range)# end
11 SW1# show interfaces status | include down
12 Fa0/10          ***UNUSED P down    1          auto    auto    10/100BaseTX
```

Security Best Practices

1. Shut down all unused ports
2. Put ports in a "dead-end" VLAN (e.g., VLAN 999)
3. Add descriptive comments
4. Document in network diagram

Verification

```
1 ! Check interface status
2 SW1# show interfaces fa0/10
3 FastEthernet0/10 is administratively down, line protocol is down
4
5 ! Check specific port security
6 SW1# show interfaces description
7 Interface          Status          Description
8 Fa0/10             administratively down ***UNUSED PORT -
   ADMIN DOWN***
```

Audit Script

```
1 ! Create audit of all unused ports
2 SW1# show run | include interface|shutdown|description
3 interface FastEthernet0/10
4   shutdown
5   description ***UNUSED PORT - ADMIN DOWN***
```

1.16 Lab 16: Configuring Banner MOTD

Objective

Configure legal notification banners for network device access.

Configuration Steps

```
1 R1> enable
2 R1# configure terminal
3
4 ! Message of the Day banner (shown before login)
5 R1(config)# banner motd #
6 Enter TEXT message. End with the character '#'.
7 *****
8 *          UNAUTHORIZED ACCESS PROHIBITED          *
9 *                                                    *
10 * This system is for authorized users only.        *
11 * All activities are monitored and recorded.        *
12 * By accessing this system, you consent            *
13 * to monitoring.                                    *
14 *****
15 #
16
17 ! Login banner (shown after MOTD, before password prompt)
18 R1(config)# banner login #
19 Enter TEXT message. End with the character '#'.
20 Authorized Personnel Only
21 #
22
23 ! Exec banner (shown after successful login)
24 R1(config)# banner exec #
25 Enter TEXT message. End with the character '#'.
26 Welcome to R1 - Production Router
27 #
```

Verification

```
1 ! Exit and reconnect
2 R1# exit
3
4 *****
5 *          UNAUTHORIZED ACCESS PROHIBITED          *
6 *                                                    *
7 * This system is for authorized users only.        *
8 * All activities are monitored and recorded.        *
9 * By accessing this system, you consent            *
10 * to monitoring.                                    *
11 *****
12
13 Authorized Personnel Only
14 User Access Verification
15 Password:
16 Welcome to R1 - Production Router
17 R1>
```

Legal Considerations

- Must specify authorized use only
- Should mention monitoring
- No misleading statements
- Clear ownership notice

1.17 Lab 17: Telnet Access to Router

Objective

Configure Telnet (unencrypted) for remote access (lab use only).

Topology

```
[PC] --- [Switch] --- [R1]
.10          .254 (vlan1)
(VTY access)
```

Configuration Steps

```
1 R1> enable
2 R1# configure terminal
3
4 ! Configure VTY lines (0-4 = 5 concurrent sessions)
5 R1(config)# line vty 0 4
6 R1(config-line)# password telnet123
7 R1(config-line)# login
8 R1(config-line)# transport input telnet
9 R1(config-line)# exec-timeout 10 0
10 R1(config-line)# logging synchronous
11 R1(config-line)# exit
12
13 ! Configure interface for management
14 R1(config)# interface vlan 1
15 R1(config-if)# ip address 192.168.1.254 255.255.255.0
16 R1(config-if)# no shutdown
17 R1(config-if)# exit
18
19 ! Enable Telnet server (default is enabled)
20 R1(config)# ip telnet server
```

Verification (from PC)

```
1 PC> telnet 192.168.1.254
2 Trying 192.168.1.254 ...Open
3 User Access Verification
4 Password: telnet123
5 R1>
```

Security Warning

- Telnet sends passwords in plain text
- Use SSH in production
- Only use Telnet in lab environments

1.18 Lab 18: SSH Configuration on Router

Objective

Configure secure SSH access for encrypted remote management.

Prerequisites

- Router must have a hostname and domain name
- Must have RSA keys generated
- Must have a username/password or AAA

Configuration Steps

```
1 R1> enable
2 R1# configure terminal
3
4 ! Step 1: Set hostname and domain name
5 R1(config)# hostname R1
6 R1(config)# ip domain-name ccnadomain.com
7
8 ! Step 2: Generate RSA keys (minimum 1024 bits for SSH v2)
9 R1(config)# crypto key generate rsa modulus 2048
10 The name for the keys will be: R1.ccnadomain.com
11 % The key modulus size is 2048 bits
12 % Generating 2048 bit RSA keys, keys will be non-exportable...[OK]
13
14 ! Step 3: Create local user
15 R1(config)# username admin secret cisco123
16
17 ! Step 4: Configure VTY lines for SSH only
18 R1(config)# line vty 0 4
19 R1(config-line)# transport input ssh
20 R1(config-line)# login local
21 R1(config-line)# exec-timeout 5 0
22 R1(config-line)# exit
23
24 ! Step 5: Enable SSH version 2 (disable version 1)
25 R1(config)# ip ssh version 2
26
27 ! Step 6: Optional - SSH timeouts
28 R1(config)# ip ssh time-out 60
29 R1(config)# ip ssh authentication-retries 3
```

Verification

```
1 ! Check SSH status
2 R1# show ip ssh
3 SSH Enabled - version 2.0
4 Authentication timeout: 60 secs; Authentication retries: 3
5
6 ! Check RSA keys
7 R1# show crypto key mypubkey rsa
```



```
8
9 ! Test from PC
10 PC> ssh -l admin 192.168.1.254
11 Password: cisco123
12 R1>
```

1.19 Lab 19: Interface Status Verification

Objective

Master interface troubleshooting using various show commands.

Interface States Explained

Status	Protocol	Meaning
up	up	Working correctly
up	down	Layer 1 OK, Layer 2 problem (encapsulation mismatch)
down	down	Cable disconnected, switch port shutdown
administratively down	down	Interface manually shutdown

Verification Commands

```

1 ! Quick overview
2 R1# show ip interface brief
3 Interface                               IP-Address      OK? Method Status Protocol
4 GigabitEthernet0/0                     192.168.1.1     YES manual up      up
5 GigabitEthernet0/1                     unassigned      YES unset  up      down
6 Serial0/0/0                             10.0.0.1        YES manual up      up
7
8 ! Detailed interface statistics
9 R1# show interfaces gigabitEthernet 0/0
10 GigabitEthernet0/0 is up, line protocol is up
11   Hardware is CN Gigabit Ethernet, address is aaaa.bbbb.cccc
12   Internet address is 192.168.1.1/24
13   MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
14   reliability 255/255, txload 1/255, rxload 1/255
15   Encapsulation ARPA, loopback not set
16   Keepalive set (10 sec)
17   Full-duplex, 100Mb/s, media type is RJ45
18   input flow-control is off, output flow-control is unsupported
19   ARP type: ARPA, ARP Timeout 04:00:00
20   Last input 00:00:00, output 00:00:00, output hang never
21   Last clearing of "show interface" counters never
22   Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
23   Queueing strategy: fifo
24   Output queue: 0/40 (size/max)
25   5 minute input rate 0 bits/sec, 0 packets/sec
26   5 minute output rate 0 bits/sec, 0 packets/sec
27     0 packets input, 0 bytes, 0 no buffer
28     Received 0 broadcasts (0 IP multicasts)
29     0 runts, 0 giants, 0 throttles
30     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
31     0 watchdog, 0 multicast, 0 pause input
32     0 packets output, 0 bytes, 0 underruns
33     0 output errors, 0 collisions, 0 interface resets
34     0 unknown protocol drops
35     0 babbles, 0 late collision, 0 deferred
36     0 lost carrier, 0 no carrier, 0 pause output
37     0 output buffer failures, 0 output buffers swapped out
38
39 ! Check interface counters

```

```
40 R1# clear counters
41 Clear "show interface" counters on all interfaces [confirm]
```

1.20 Lab 20: Basic Troubleshooting (Layer 1 & 2)

Objective

Systematically troubleshoot physical and data link layer issues.

Troubleshooting Methodology

1. Identify the problem
2. Establish theory of probable cause
3. Test the theory
4. Establish action plan
5. Implement solution
6. Verify full functionality
7. Document findings

Common Layer 1 Issues

Issue	Symptom	Solution
Wrong cable type	No link light	Use straight-through for PC-Switch, crossover for switch-switch
Distance exceeded	Intermittent connectivity	Max 100m for copper, use fiber for longer distances
Faulty cable	CRC errors	Replace cable
Bad port	Interface down/down	Move to different port
Power issues	Device unreachable	Check power supply

Common Layer 2 Issues

Issue	Symptom	Solution
Duplex mismatch	High collisions	Set both ends to auto or same manual setting
VLAN mismatch	No connectivity	Verify both ends in same VLAN
Trunk issues	Intermittent traffic	Check native VLAN matching
MAC flooding	Network slow	Implement port security
STP blockage	No traffic flow	Verify root bridge placement

Troubleshooting Commands

```
1 ! Layer 1 checks
2 R1# show interfaces | include line protocol
3 R1# show interfaces status
4 R1# show logging
5
6 ! Layer 2 checks
7 SW1# show vlan brief
8 SW1# show mac address-table
9 SW1# show spanning-tree
10 SW1# show interfaces trunk
11
12 ! Cable testing (if supported)
13 SW1# test cable-diagnostics tdr interface fa0/1
14 SW1# show cable-diagnostics tdr interface fa0/1
```

Ping Troubleshooting Flowchart

```
Start: ping fails
|
v
Check IP configuration -> Wrong IP? -> Fix IP
|
v (correct)
Check default gateway -> Missing? -> Add gateway
|
v (present)
Check switch port status -> Down? -> Enable port
|
v (up)
Check VLAN assignment -> Wrong VLAN? -> Reassign
|
v (correct)
Check ACLs -> Blocking? -> Modify ACL
|
v (no ACL)
Check routing -> No route? -> Add route
```

1.21 Lab 21: Password Recovery Basics

Objective

Perform password recovery on Cisco routers and switches.

Important Note

This is for educational purposes only. Only perform on devices you own.

Router Password Recovery Process

1. Power cycle the router

2. Press Ctrl+Break during boot (within 60 seconds)

```
1 rommon 1 >  
2
```

3. Change configuration register to ignore startup config

```
1 rommon 1 > confreg 0x2142  
2 rommon 2 > reset  
3
```

4. After reboot, enter privileged mode (no password)

```
1 Router> enable  
2 Router#  
3
```

5. Load startup config into running config

```
1 Router# copy startup-config running-config  
2 Destination filename [running-config]?  
3
```

6. Change passwords

```
1 Router# configure terminal  
2 Router(config)# enable secret newpassword  
3 Router(config)# line console 0  
4 Router(config-line)# password newpassword  
5 Router(config-line)# login  
6 Router(config-line)# exit  
7 Router(config)# line vty 0 4  
8 Router(config-line)# password newpassword  
9 Router(config-line)# login  
10 Router(config-line)# exit  
11
```

7. Reset configuration register back to normal

```
1 Router(config)# config-register 0x2102  
2 Router(config)# end  
3 Router# copy running-config startup-config  
4 Router# reload  
5
```

Configuration Register Values

Value	Meaning
0x2102	Normal (load startup config)
0x2142	Ignore startup config
0x2101	Boot into ROM monitor

Switch Password Recovery (Different Process)

1. Power cycle switch
2. Hold "Mode" button for 15 seconds
3. Enter switch: prompt

```
1 switch:
2
```

4. Initialize flash

```
1 switch: flash_init
2 switch: load_helper
3
```

5. Rename config file

```
1 switch: rename flash:config.text flash:config.old
2 switch: boot
3
```

6. Recover configuration

```
1 Switch# rename flash:config.old flash:config.text
2 Switch# copy flash:config.text running-config
3
```

1.22 Lab 22: Using Help and Command Shortcuts

Objective

Master Cisco CLI productivity features.

Context-Sensitive Help

```

1 ! List all commands starting with "sh"
2 R1# sh?
3 show  shutdown
4
5 ! List commands starting with "show ip"
6 R1# show ip?
7   access-list  arp  bgp  cache  dhcp  eigrp  flow  http  interface
8   nat  ospf  protocols  route  sla  ssh  traffic
9
10 ! Show command syntax
11 R1# show ?
12 R1# show ip ?
13 R1# show ip route ?
14
15 ! Partial keyword help
16 R1# show ip ro?
17   route

```

Command Shortcuts

Shortcut	Full Command	Use
en	enable	Enter privileged mode
conf t	configure terminal	Global config mode
int	interface	Interface config mode
sh	show	Display information
sh ip int br	show ip interface brief	Quick interface status
wr	write memory	Save config

Command History

```

1 ! Show command history
2 R1# show history
3
4 ! Increase history buffer
5 R1# terminal history size 100
6
7 ! Recall previous commands
8 Ctrl+P or Up Arrow - Previous command
9 Ctrl+N or Down Arrow - Next command
10
11 ! Search history
12 Ctrl+R - Reverse search (type part of command)

```


Editing Shortcuts

Keystroke	Action
Ctrl+A	Move to beginning of line
Ctrl+E	Move to end of line
Ctrl+F or Right Arrow	Forward one character
Ctrl+B or Left Arrow	Back one character
Ctrl+D	Delete character at cursor
Ctrl+K	Delete from cursor to end of line
Ctrl+U	Delete entire line
Ctrl+W	Delete previous word
Tab	Complete command/parameter
?	Context-sensitive help

Command Aliases

```

1 ! Create custom shortcuts
2 R1(config)# alias exec sb show ip interface brief
3 R1(config)# alias exec sr show running-config
4 R1(config)# alias exec sc show clock
5
6 ! Use alias
7 R1# sb
8 Interface                IP-Address      OK? Method Status Protocol
9 GigabitEthernet0/0       192.168.1.1     YES manual up      up
10
11 ! Remove alias
12 R1(config)# no alias exec sb

```

1.23 Lab 23: Configuring Clock on Router

Objective

Set system clock and timezone for accurate logging and timestamps.

Configuration Steps

```
1 R1> enable
2 R1# configure terminal
3
4 ! Manual clock setting (24-hour format)
5 R1(config)# clock timezone EST -5
6 R1(config)# clock summer-time EDT recurring
7 R1(config)# exit
8 R1# clock set 14:30:00 15 Mar 2024
9
10 ! Alternative: Set calendar (hardware clock)
11 R1# calendar set 14:30:00 15 Mar 2024
12
13 ! Sync calendar to system clock
14 R1# clock read-calendar
```

Verification

```
1 ! Show current time
2 R1# show clock
3 14:30:15.123 EST Thu Mar 15 2024
4
5 ! Show with milliseconds
6 R1# show clock detail
7 14:30:15.123 EST Thu Mar 15 2024
8 Time source is user configuration
9
10 ! Show calendar (hardware clock)
11 R1# show calendar
12 14:30:20 EST Thu Mar 15 2024
```

Enable Timestamps

```
1 R1(config)# service timestamps debug datetime msec localtime show-
   timezone
2 R1(config)# service timestamps log datetime msec localtime show-
   timezone
3
4 ! Now log messages will show accurate time
5 R1# show logging
6 *Mar 15 14:30:15.123 EST: %SYS-5-CONFIG_I: Configured from console
```

NTP Integration (Preview of Lab 58)

```
1 ! Use NTP for automatic time sync
2 R1(config)# ntp server 0.pool.ntp.org
3 R1# show ntp status
4 Clock is synchronized, stratum 2
```

1.24 Lab 24: Backup Configuration via TFTP

Objective

Backup device configurations to a TFTP server for disaster recovery.

Topology

```
[R1] --- [Switch] --- [TFTP Server]
.1          (192.168.1.100)
```

Prerequisites

- TFTP server configured (use PC with TFTP software)
- IP connectivity between router and TFTP server

Configuration Steps

```
1 ! Verify connectivity to TFTP server
2 R1# ping 192.168.1.100
3 !!!!!
4
5 ! Backup running-config
6 R1# copy running-config tftp
7 Address or name of remote host []? 192.168.1.100
8 Destination filename [R1-config]? R1-running-config-2024-03-15
9 !!
10 [OK - 1245 bytes]
11
12 ! Backup startup-config
13 R1# copy startup-config tftp
14 Address or name of remote host []? 192.168.1.100
15 Destination filename [R1-config]? R1-startup-config
16 !!
17 [OK - 1245 bytes]
18
19 ! Backup entire flash (IOS image)
20 R1# copy flash: tftp
21 Source filename []? c2900-universalk9-mz.SPA.157-3.M3.bin
22 Address or name of remote host []? 192.168.1.100
23 Destination filename [c2900-universalk9-mz.SPA.157-3.M3.bin]?
24 !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
25 [OK - 45875200 bytes]
```

Verification on TFTP Server

```
1 ! On PC (TFTP server)
2 TFTP Server Log:
3 Received R1-running-config-2024-03-15 (1245 bytes)
4 Received R1-startup-config (1245 bytes)
5 Received c2900-universalk9-mz.SPA.157-3.M3.bin (45875200 bytes)
```

Multiple Device Backup Script

```
1 ! On management PC (Windows batch file)
2 @echo off
3 echo Backing up R1...
4 tftp -i 192.168.1.1 PUT running-config R1-config.txt
5 echo Backing up SW1...
6 tftp -i 192.168.1.2 PUT running-config SW1-config.txt
7 echo Backup complete!
```

1.25 Lab 25: Restore Configuration from TFTP

Objective

Restore device configuration from TFTP backup after failure.

Restore Process

```
1 ! Method 1: Merge configuration (adds to existing)
2 R1# copy tftp running-config
3 Address or name of remote host []? 192.168.1.100
4 Source filename []? R1-running-config-2024-03-15
5 Destination filename [running-config]?
6 !!
7 [OK - 1245 bytes]
8
9 ! Method 2: Replace entire configuration
10 ! First, erase current config
11 R1# write erase
12 Erasing the nvram filesystem will remove all configuration files!
13   Continue? [confirm]
14 [OK]
15 R1# reload
16
17 ! After reboot, restore
18 R1# copy tftp startup-config
19 Address or name of remote host []? 192.168.1.100
20 Source filename []? R1-startup-config
21 Destination filename [startup-config]?
22 !!
23 [OK - 1245 bytes]
24
25 ! Reload to load restored config
26 R1# reload
```

Restore IOS Image from TFTP

```
1 ! If router is in ROMmon mode
2 rommon 1 > IP_ADDRESS=192.168.1.1
3 rommon 2 > IP_SUBNET_MASK=255.255.255.0
4 rommon 3 > DEFAULT_GATEWAY=192.168.1.254
5 rommon 4 > TFTP_SERVER=192.168.1.100
6 rommon 5 > TFTP_FILE=c2900-universalk9-mz.SPA.157-3.M3.bin
7 rommon 6 > tftpdnld
8
9 ! Or from IOS
10 R1# copy tftp flash:
11 Address or name of remote host []? 192.168.1.100
12 Source filename []? c2900-universalk9-mz.SPA.157-3.M3.bin
13 Destination filename [c2900-universalk9-mz.SPA.157-3.M3.bin]?
14 !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

Verification

```
1 ! Verify restored configuration
2 R1# show running-config
3 R1# show version
```

2 Intermediate Level: VLANs, Routing & Services

Week 3-5: Switching Technologies

2.1 Lab 26: Creating VLANs

Objective

Create VLANs to segment Layer 2 broadcast domains.

Topology

[PC1-VLAN10] --- [SW1] --- [PC2-VLAN20]

Configuration Steps

```
1 SW1> enable
2 SW1# configure terminal
3
4 ! Create VLAN 10
5 SW1(config)# vlan 10
6 SW1(config-vlan)# name Sales
7 SW1(config-vlan)# exit
8
9 ! Create VLAN 20
10 SW1(config)# vlan 20
11 SW1(config-vlan)# name Engineering
12 SW1(config-vlan)# exit
13
14 ! Create VLAN 99 (Management)
15 SW1(config)# vlan 99
16 SW1(config-vlan)# name Management
17 SW1(config-vlan)# exit
18
19 ! Assign ports to VLANs
20 SW1(config)# interface fastEthernet 0/1
21 SW1(config-if)# switchport mode access
22 SW1(config-if)# switchport access vlan 10
23 SW1(config-if)# exit
24
25 SW1(config)# interface fastEthernet 0/2
26 SW1(config-if)# switchport mode access
27 SW1(config-if)# switchport access vlan 20
28 SW1(config-if)# end
29
30 ! Configure VLAN interface (SVI) for management
31 SW1(config)# interface vlan 99
32 SW1(config-if)# ip address 192.168.99.2 255.255.255.0
33 SW1(config-if)# no shutdown
34 SW1(config-if)# exit
35 SW1(config)# ip default-gateway 192.168.99.1
```

Verification


```
1 ! Show VLANs
2 SW1# show vlan brief
3 VLAN Name                                Status      Ports
4 ----
5 1      default                            active      Fa0/3, Fa0/4, Fa0/5,
6      Fa0/6
7 10     Sales                             active      Fa0/1
8 20     Engineering                       active      Fa0/2
9 99     Management                        active
10
10 ! Show VLAN information
11 SW1# show vlan id 10
12 VLAN ID: 10
13 Name: Sales
14 State: active
15
16 ! Test isolation (PC1 cannot ping PC2 even with same subnet)
17 PC1> ping 192.168.1.20
18 Request timed out.
```

2.2 Lab 27: Assigning Ports to VLANs

Objective

Learn multiple methods for assigning switch ports to VLANs.

Configuration Methods

```
1 ! Method 1: Single port
2 SW1(config)# interface fastEthernet 0/10
3 SW1(config-if)# switchport mode access
4 SW1(config-if)# switchport access vlan 30
5 SW1(config-if)# exit
6
7 ! Method 2: Range of ports
8 SW1(config)# interface range fastEthernet 0/11-20
9 SW1(config-if-range)# switchport mode access
10 SW1(config-if-range)# switchport access vlan 40
11 SW1(config-if-range)# exit
12
13 ! Method 3: Interface range with mixed types
14 SW1(config)# interface range gigabitEthernet 0/1-2, fastEthernet 0/24
15 SW1(config-if-range)# switchport mode access
16 SW1(config-if-range)# switchport access vlan 50
17
18 ! Method 4: VLAN database mode (older IOS)
19 SW1# vlan database
20 SW1(vlan)# vlan 60 name Marketing
21 SW1(vlan)# apply
22 SW1(vlan)# exit
```

Verification

```
1 ! Verify port assignments
2 SW1# show interfaces status
3 Port      Name      Status      Vlan      Duplex  Speed  Type
4 Fa0/1      connected  10          full      100     10/100BaseTX
5 Fa0/2      connected  20          full      100     10/100BaseTX
6 Fa0/10     notconnect 30          auto      auto    10/100BaseTX
7 Fa0/11-20  notconnect 40          auto      auto    10/100BaseTX
8
9 ! Detailed port VLAN info
10 SW1# show interfaces fa0/1 switchport
11 Name: Fa0/1
12 Switchport: Enabled
13 Administrative Mode: static access
14 Operational Mode: static access
15 Administrative Trunking Encapsulation: dot1q
16 Operational Trunking Encapsulation: native
17 Negotiation of Trunking: Off
18 Access Mode VLAN: 10 (Sales)
19 Trunking Native Mode VLAN: 1 (default)
```

Best Practices

- Use descriptive VLAN names

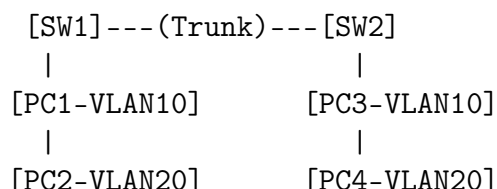
- Document port assignments
- Use interface ranges for efficiency
- Keep management VLAN separate

2.3 Lab 28: VLAN Trunking (802.1Q)

Objective

Configure 802.1Q trunks to carry multiple VLANs between switches.

Topology



Configuration Steps

```

1 ! On SW1
2 SW1> enable
3 SW1# configure terminal
4 SW1(config)# interface gigabitEthernet 0/1
5 SW1(config-if)# switchport trunk encapsulation dot1q
6 SW1(config-if)# switchport mode trunk
7 SW1(config-if)# switchport trunk native vlan 99
8 SW1(config-if)# switchport trunk allowed vlan 10,20,99
9 SW1(config-if)# end
10
11 ! On SW2 (same configuration)
12 SW2> enable
13 SW2# configure terminal
14 SW2(config)# interface gigabitEthernet 0/1
15 SW2(config-if)# switchport trunk encapsulation dot1q
16 SW2(config-if)# switchport mode trunk
17 SW2(config-if)# switchport trunk native vlan 99
18 SW2(config-if)# switchport trunk allowed vlan 10,20,99
19 SW2(config-if)# end
  
```

Verification

```

1 ! Verify trunk status
2 SW1# show interfaces trunk
3 Port      Mode      Encapsulation  Status      Native vlan
4 Gi0/1     on        802.1q         trunking    99
5
6 Port      Vlans allowed on trunk
7 Gi0/1     10,20,99
8
9 Port      Vlans allowed and active in management domain
10 Gi0/1    10,20,99
11
12 Port      Vlans in spanning tree forwarding state and not pruned
13 Gi0/1    10,20,99
14
15 ! Verify allowed VLANs
16 SW1# show interfaces gigabitEthernet 0/1 switchport
17 Trunking Native Mode VLAN: 99 (Management)
  
```

```
18 Trunking VLANs Enabled: 10,20,99
19 Pruning VLANs Enabled: 2-1001
20
21 ! Test connectivity across trunk
22 PC1 (VLAN10) ping PC3 (VLAN10) -> Successful
23 PC1 (VLAN10) ping PC4 (VLAN20) -> Fails (different VLANs)
```

2.4 Lab 29: Native VLAN Configuration

Objective

Understand and configure the native VLAN for untagged traffic on trunks.

Key Concepts

- Native VLAN carries untagged traffic on trunk
- Default native VLAN is VLAN 1
- Both ends must match to prevent VLAN hopping

Configuration Steps

```
1 ! Change native VLAN on SW1
2 SW1(config)# interface gigabitEthernet 0/1
3 SW1(config-if)# switchport trunk native vlan 999
4 SW1(config-if)# end
5
6 ! Change native VLAN on SW2 (MUST match)
7 SW2(config)# interface gigabitEthernet 0/1
8 SW2(config-if)# switchport trunk native vlan 999
9 SW2(config-if)# end
10
11 ! Verify native VLAN
12 SW1# show interfaces trunk
13 Port          Mode          Encapsulation  Status        Native vlan
14 Gi0/1         on            802.1q         trunking      999
15
16 ! Create isolated VLAN for native (security best practice)
17 SW1(config)# vlan 999
18 SW1(config-vlan)# name NATIVE_UNUSED
19 SW1(config-vlan)# exit
20
21 ! Remove VLAN 1 from trunk allowed list
22 SW1(config)# interface gigabitEthernet 0/1
23 SW1(config-if)# switchport trunk allowed vlan remove 1
```

Security Issue: VLAN Hopping

```
1 ! Misconfiguration that allows attack
2 ! SW1: native vlan 1, SW2: native vlan 999
3 ! Attacker can send double-tagged 802.1Q frames
4
5 ! Prevention:
6 ! 1. Use unused VLAN as native
7 ! 2. Disable DTP (Dynamic Trunking Protocol)
8 SW1(config)# interface gigabitEthernet 0/1
9 SW1(config-if)# switchport nonegotiate
10 ! 3. Explicitly set trunk mode
11 SW1(config-if)# switchport mode trunk
```

2.5 Lab 30: Inter-VLAN Routing (Router-on-a-Stick)

Objective

Route between VLANs using a single router interface with subinterfaces.

Topology

```
[PC1-VLAN10] --- [SW1] --- (Trunk) --- [R1]
[PC2-VLAN20] --- [SW1]                (.1)
192.168.10.0/24          192.168.20.0/24
```

Configuration Steps

```
1 ! On SW1 - Trunk to router
2 SW1(config)# interface gigabitEthernet 0/1
3 SW1(config-if)# switchport mode trunk
4 SW1(config-if)# switchport trunk native vlan 99
5 SW1(config-if)# switchport trunk allowed vlan 10,20,99
6
7 ! On R1 - Subinterfaces
8 R1> enable
9 R1# configure terminal
10 R1(config)# interface gigabitEthernet 0/0.10
11 R1(config-subif)# encapsulation dot1Q 10
12 R1(config-subif)# ip address 192.168.10.1 255.255.255.0
13 R1(config-subif)# exit
14
15 R1(config)# interface gigabitEthernet 0/0.20
16 R1(config-subif)# encapsulation dot1Q 20
17 R1(config-subif)# ip address 192.168.20.1 255.255.255.0
18 R1(config-subif)# exit
19
20 R1(config)# interface gigabitEthernet 0/0.99
21 R1(config-subif)# encapsulation dot1Q 99 native
22 R1(config-subif)# ip address 192.168.99.1 255.255.255.0
23 R1(config-subif)# exit
24
25 ! Enable physical interface
26 R1(config)# interface gigabitEthernet 0/0
27 R1(config-if)# no shutdown
```

Verification

```
1 ! On R1 - Verify subinterfaces
2 R1# show ip interface brief
3 GigabitEthernet0/0          unassigned      YES unset   up        up
4 GigabitEthernet0/0.10      192.168.10.1  YES manual  up        up
5 GigabitEthernet0/0.20      192.168.20.1  YES manual  up        up
6 GigabitEthernet0/0.99      192.168.99.1  YES manual  up        up
7
8 ! On PC1 (VLAN10)
9 PC1> ipconfig 192.168.10.10 255.255.255.0 192.168.10.1
10
11 ! On PC2 (VLAN20)
```

```
12 PC2> ipconfig 192.168.20.20 255.255.255.0 192.168.20.1
13
14 ! Test inter-VLAN routing
15 PC1> ping 192.168.20.20
16 Reply from 192.168.20.20: bytes=32 time=1ms TTL=127
```


2.6 Lab 31: VLAN Troubleshooting

Objective

Systematically troubleshoot common VLAN and trunking issues.

Common Issues and Solutions

Issue	Solution
VLAN mismatch	Verify access VLAN on both sides: <code>show interfaces switchport</code>
Trunk not forming	Check encapsulation: <code>switchport trunk encapsulation dot1q</code>
Native VLAN mismatch	Verify native VLAN matches: <code>show interfaces trunk</code>
Missing VLAN on trunk	Check allowed VLAN list: <code>switchport trunk allowed vlan</code>
Wrong IP subnet	Verify PCs have correct gateway for their VLAN
STP blocking	Check spanning-tree: <code>show spanning-tree vlan 10</code>

Troubleshooting Commands

```

1 ! Step 1: Verify VLAN existence
2 SW1# show vlan brief
3
4 ! Step 2: Verify port assignments
5 SW1# show interfaces status
6 SW1# show interfaces fa0/1 switchport
7
8 ! Step 3: Verify trunk
9 SW1# show interfaces trunk
10 SW1# show interfaces gi0/1 switchport
11
12 ! Step 4: Verify MAC addresses
13 SW1# show mac address-table | include 10
14
15 ! Step 5: Verify router subinterfaces
16 R1# show ip route
17 R1# show interfaces | include line protocol|encapsulation
18
19 ! Step 6: Verify PC configuration
20 PC> ipconfig
21 PC> arp -a
22
23 ! Step 7: Check for VLAN pruning
24 SW1# show interfaces trunk | include Pruning
25
26 ! Debug commands (use cautiously)
27 SW1# debug trunk
28 SW1# debug vlan packet

```

Troubleshooting Flowchart

PC cannot ping gateway

|

v

Check PC IP config -> Wrong? -> Fix

|

v (correct)

Check switchport VLAN -> Wrong? -> Fix

|

v (correct)

Check trunk between switch and router -> Down? -> Fix

|

v (up)

Check router subinterface config -> Missing/error? -> Fix

|

v (correct)

Check router routing table -> Route exists? -> Add

2.7 Lab 32: VTP Configuration (Server/Client)

Objective

Configure VLAN Trunking Protocol to manage VLANs across switches.

Topology

[SW1]---(Trunk)---[SW2]---(Trunk)---[SW3]
(VTP Server) (VTP Client) (VTP Client)

Configuration Steps

```
1 ! On SW1 (VTP Server)
2 SW1(config)# vtp mode server
3 SW1(config)# vtp domain CCNA
4 SW1(config)# vtp password cisco123
5 SW1(config)# vtp version 2
6 SW1(config)# end
7 SW1# show vtp status
8
9 ! On SW2 (VTP Client)
10 SW2(config)# vtp mode client
11 SW2(config)# vtp domain CCNA
12 SW2(config)# vtp password cisco123
13 SW2(config)# vtp version 2
14
15 ! On SW3 (VTP Client)
16 SW3(config)# vtp mode client
17 SW3(config)# vtp domain CCNA
18 SW3(config)# vtp password cisco123
19 SW3(config)# vtp version 2
20
21 ! Now create VLAN on Server (SW1)
22 SW1(config)# vlan 100
23 SW1(config-vlan)# name Sales
24 SW1(config-vlan)# exit
25 SW1(config)# vlan 200
26 SW1(config-vlan)# name Engineering
```

Verification

```
1 ! On SW2 (should see VLANs automatically)
2 SW2# show vlan brief
3 100 Sales active
4 200 Engineering active
5
6 ! VTP status
7 SW2# show vtp status
8 VTP Version : 2
9 Configuration Revision : 5
10 Maximum VLANs supported locally : 255
11 Number of existing VLANs : 10
12 VTP Operating Mode : Client
13 VTP Domain Name : CCNA
```

```
14 VTP Pruning Mode           : Disabled
15 VTP V2 Mode               : Enabled
16 VTP Traps Generation      : Disabled
17 MD5 digest                 : 0x7A 0x3B 0x4C 0x1D
18 Configuration last modified by 192.168.1.1 at 3-15-24 14:30:00
19
20 ! Verify VTP advertisements
21 SW1# show vtp counters
```

VTP Versions

Version	Features
---------	----------

- | | |
|---|---|
| 1 | Basic VLAN propagation |
| 2 | Token Ring support, consistency checks |
| 3 | Extended VLANs (1006-4094), private VLANs |

2.8 Lab 33: VTP Pruning

Objective

Configure VTP pruning to reduce unnecessary broadcast traffic on trunks.

Configuration Steps

```
1 ! Enable VTP pruning on VTP Server
2 SW1(config)# vtp pruning
3
4 ! Verify pruning
5 SW1# show vtp status
6 VTP Pruning Mode           : Enabled
7
8 ! Check pruned VLANs on trunk
9 SW1# show interfaces trunk
10 Port          Vlans in spanning tree forwarding state and not pruned
11 Gi0/1         1,100,200
12
13 ! Note: VLANs not present on downstream switches are pruned
```

Pruning Benefits

- Reduces broadcast traffic on trunks
- Improves bandwidth utilization
- Only sends VLAN traffic where needed

Manual Pruning (without VTP)

```
1 ! Manually prune VLANs on trunk
2 SW1(config)# interface gigabitEthernet 0/1
3 SW1(config-if)# switchport trunk allowed vlan remove 300-400
4 SW1(config-if)# switchport trunk allowed vlan add 100-200
```

2.9 Lab 34: Private VLAN Basics

Objective

Configure Private VLANs for isolation within the same VLAN.

PVLAN Components

- **Primary VLAN:** Main VLAN
- **Secondary VLANs:** Community or Isolated
- **Promiscuous Port:** Gateway/router connection
- **Isolated Port:** Can only talk to promiscuous ports
- **Community Port:** Can talk within community + promiscuous

Configuration Steps

```
1 ! Enable private VLAN feature
2 SW1(config)# vlan 100
3 SW1(config-vlan)# private-vlan primary
4 SW1(config-vlan)# exit
5
6 ! Create secondary VLANs
7 SW1(config)# vlan 101
8 SW1(config-vlan)# private-vlan isolated
9 SW1(config-vlan)# exit
10
11 SW1(config)# vlan 102
12 SW1(config-vlan)# private-vlan community
13 SW1(config-vlan)# exit
14
15 ! Associate secondary with primary
16 SW1(config)# vlan 100
17 SW1(config-vlan)# private-vlan association 101,102
18 SW1(config-vlan)# exit
19
20 ! Configure ports
21 ! Promiscuous port (to router)
22 SW1(config)# interface gigabitEthernet 0/1
23 SW1(config-if)# switchport mode private-vlan promiscuous
24 SW1(config-if)# switchport private-vlan mapping 100 101,102
25
26 ! Isolated port
27 SW1(config)# interface fastEthernet 0/10
28 SW1(config-if)# switchport mode private-vlan host
29 SW1(config-if)# switchport private-vlan host-association 100 101
30
31 ! Community port
32 SW1(config)# interface fastEthernet 0/20
33 SW1(config-if)# switchport mode private-vlan host
34 SW1(config-if)# switchport private-vlan host-association 100 102
```

Verification

```
1 SW1# show vlan private-vlan
2 Primary Secondary Type          Ports
3 100      101      isolated      Fa0/10
4 100      102      community     Fa0/20
5
6 SW1# show interfaces private-vlan mapping
7 Interface      Primary VLAN      Secondary VLAN      Type
8 Gi0/1          100              101,102            promiscuous
9 Fa0/10         100              101                 host (isolated)
10 Fa0/20         100              102                 host (community)
```

2.10 Lab 35: Voice VLAN Setup

Objective

Configure Voice VLAN for IP phones with QoS marking.

Topology

```
[IP Phone] --- [SW1] --- [Router]
      |
      | (Voice VLAN 150)
[PC]   (Data VLAN 10)
```

Configuration Steps

```
1 ! Create VLANs
2 SW1(config)# vlan 10
3 SW1(config-vlan)# name DATA
4 SW1(config-vlan)# exit
5
6 SW1(config)# vlan 150
7 SW1(config-vlan)# name VOICE
8 SW1(config-vlan)# exit
9
10 ! Configure port for voice and data
11 SW1(config)# interface fastEthernet 0/1
12 SW1(config-if)# switchport mode access
13 SW1(config-if)# switchport access vlan 10
14 SW1(config-if)# switchport voice vlan 150
15 SW1(config-if)# mls qos trust cos
16 SW1(config-if)# auto qos voip cisco-phone
17 SW1(config-if)# spanning-tree portfast
18 SW1(config-if)# end
```

Verification

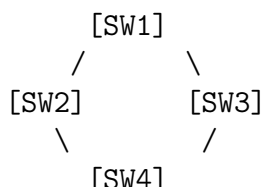
```
1 ! Show voice VLAN
2 SW1# show interfaces fa0/1 switchport | include Voice
3 Voice VLAN: 150 (VOICE)
4
5 ! Show QoS
6 SW1# show mls qos interface fa0/1
7 FastEthernet0/1
8 trust state: trust cos
9 trust mode: trust cos
10 COS override: dis
11 default COS: 0
12
13 ! CDP shows phone
14 SW1# show cdp neighbors fa0/1
15 Device ID: SEP123456789ABC
16 Interface: Fa0/1, Port ID: Port 1
17 Platform: Cisco IP Phone 7960
```


2.11 Lab 36: Spanning Tree Protocol (STP) Basics

Objective

Understand STP operation and view STP status.

Topology



Configuration Steps (No config needed - STP default)

```

1 ! View root bridge
2 SW1# show spanning-tree
3 VLAN0001
4   Spanning tree enabled protocol ieee
5   Root ID      Priority      32769
6               Address        0001.1234.5678
7               This bridge is the root
8               Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec
9
10  Bridge ID     Priority      32769 (priority 32768 sys-id-ext 1)
11              Address        0001.1234.5678
12              Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec
13              Aging Time   300 sec
14
15 Interface      Role Sts Cost          Prio.Nbr Type
16 -----
17 Gi0/1          Desg FWD 4             128.1    P2p
18 Gi0/2          Desg FWD 4             128.2    P2p
19 Fa0/1          Altn BLK 19          128.3    P2p
20
21 ! View port roles
22 SW1# show spanning-tree interface fastEthernet 0/1
23 Vlan           Role Sts Cost          Prio.Nbr Type
24 -----
25 VLAN0001       Altn BLK 19          128.3    P2p
  
```

STP Port States

State	Description
Blocking	No BPDU received, no data
Listening	Receiving BPDUs, no data (15 sec)
Learning	Building MAC table, no data (15 sec)
Forwarding	Normal operation
Disabled	Administratively down

2.12 Lab 37: Rapid STP Configuration

Objective

Configure Rapid Spanning Tree Protocol (802.1w) for faster convergence.

Configuration Steps

```
1 ! Enable RSTP globally
2 SW1(config)# spanning-tree mode rapid-pvst
3 SW2(config)# spanning-tree mode rapid-pvst
4 SW3(config)# spanning-tree mode rapid-pvst
5
6 ! Verify
7 SW1# show spanning-tree summary
8 Switch is in rapid-pvst mode
9 Root bridge for: VLAN0001
10 Extended system ID is enabled
11 Portfast Default is disabled
12
13 ! Convergence comparison
14 ! STP: 30-50 seconds
15 ! RSTP: 1-6 seconds
16
17 ! Configure edge ports (Portfast equivalent)
18 SW1(config)# interface fastEthernet 0/1
19 SW1(config-if)# spanning-tree portfast edge
20 SW1(config-if)# spanning-tree bpduguard enable
```

RSTP Port Roles

Role	Description
Root	Forwarding toward root bridge
Designated	Forwarding away from root
Alternate	Backup to root port (blocking)
Backup	Backup to designated port
Edge	Portfast enabled (host facing)

2.13 Lab 38: Root Bridge Election

Objective

Control root bridge placement for optimal traffic flow.

Configuration Steps

```
1 ! Method 1: Set priority (lower value wins)
2 SW1(config)# spanning-tree vlan 1 priority 4096
3
4 ! Method 2: Use root primary macro
5 SW1(config)# spanning-tree vlan 1 root primary
6 ! Sets priority to 24576 (or 4096 less than current root)
7
8 ! Method 3: Use root secondary macro
9 SW2(config)# spanning-tree vlan 1 root secondary
10 ! Sets priority to 28672
11
12 ! Verify root bridge
13 SW1# show spanning-tree vlan 1 | include "This bridge is the root"
14 This bridge is the root
15
16 ! Set priority for specific VLANs
17 SW1(config)# spanning-tree vlan 10 priority 4096
18 SW1(config)# spanning-tree vlan 20 priority 8192
19 SW1(config)# spanning-tree vlan 30 priority 12288
20
21 ! Verify per-VLAN root
22 SW1# show spanning-tree vlan 10
23 SW1# show spanning-tree vlan 20
```

Priority Values

Priority	Usage
0	Highest (never used - conflicts with bridge ID)
4096	Primary root (use <code>root primary</code>)
8192	Secondary root
12288	Tertiary
32768	Default (all switches)
61440	Lowest possible

Best Practices

- Set root bridge on distribution/core switches
- Set secondary root on backup distribution switch
- Use per-VLAN root for load balancing
- Never use priority 0

2.14 Lab 39: PortFast & BPDU Guard

Objective

Configure PortFast for host ports and BPDU Guard for security.

Configuration Steps

```
1 ! Configure PortFast on access ports
2 SW1(config)# interface range fastEthernet 0/1-24
3 SW1(config-if-range)# spanning-tree portfast
4 %Warning: portfast should only be enabled on ports connected to a
   single host.
5 SW1(config-if-range)# spanning-tree bpduguard enable
6
7 ! Alternative: Global PortFast for all access ports
8 SW1(config)# spanning-tree portfast default
9 SW1(config)# spanning-tree portfast bpduguard default
10
11 ! Verify PortFast
12 SW1# show spanning-tree interface fa0/1 detail
13 Port 3 (FastEthernet0/1) of VLAN0001 is forwarding
14 Port path cost 19, Port priority 128, Port Identifier 128.3.
15 Designated root has priority 32769, address 0001.1234.5678
16 Designated bridge has priority 32769, address 0001.1234.5678
17 Designated port id is 128.3, designated path cost 0
18 Timers: message age 0, forward delay 0, hold 0
19 Number of transitions to forwarding state: 1
20 Link type is point-to-point by default
21 **PortFast is configured (portfast configured)**
22
23 ! BPDU Guard action (if BPDU received)
24 SW1# show spanning-tree summary
25 Portfast BPDU Guard is enabled globally
```

PortFast Benefits

- Bypasses listening/learning states
- Immediate transition to forwarding
- Saves 30 seconds per port
- Safe only on host-facing ports

BPDU Guard Behavior

- Shuts down port if BPDU received
- Prevents rogue switch connections
- Port enters err-disabled state

Recovering from err-disabled

```
1 ! Manual recovery
2 SW1(config)# interface fastEthernet 0/1
3 SW1(config-if)# shutdown
4 SW1(config-if)# no shutdown
5
6 ! Automatic recovery
7 SW1(config)# errdisable recovery cause bpduguard
8 SW1(config)# errdisable recovery interval 300
```

2.15 Lab 40: EtherChannel (LACP)

Objective

Configure Link Aggregation Control Protocol for bandwidth and redundancy.

Topology

[SW1]===== [SW2]
(2 or 4 port bundle)

Configuration Steps

```

1 ! On SW1
2 SW1(config)# interface range gigabitEthernet 0/1-2
3 SW1(config-if-range)# channel-group 1 mode active
4 SW1(config-if-range)# exit
5 SW1(config)# interface port-channel 1
6 SW1(config-if)# switchport mode trunk
7 SW1(config-if)# switchport trunk allowed vlan 10,20,30
8
9 ! On SW2
10 SW2(config)# interface range gigabitEthernet 0/1-2
11 SW2(config-if-range)# channel-group 1 mode active
12 SW2(config-if-range)# exit
13 SW2(config)# interface port-channel 1
14 SW2(config-if)# switchport mode trunk
15 SW2(config-if)# switchport trunk allowed vlan 10,20,30

```

LACP Modes

Mode	Description
Active	Actively negotiates (sends LACP packets)
Passive	Responds to negotiations (doesn't initiate)
On	Static configuration (no negotiation)

Verification

```

1 ! Show EtherChannel summary
2 SW1# show etherchannel summary
3 Flags:  D - down          P - bundled in port-channel
4         I - stand-alone  s - suspended
5         H - Hot-standby (LACP only)
6         R - Layer3       S - Layer2
7         U - in use       f - failed to allocate aggregator
8
9 Number of channel-groups in use: 1
10 Number of aggregators:          1
11
12 Group  Port-channel  Protocol    Ports
13 -----+-----+-----+-----
14 1      Po1(SU)        LACP        Gi0/1(P)   Gi0/2(P)
15

```

```
16 ! Show port-channel details
17 SW1# show etherchannel 1 detail
18 ! Show load balancing
19 SW1# show etherchannel load-balance
20 EtherChannel Load-Balancing Configuration:
21     src-mac
22
23 ! Configure load balancing
24 SW1(config)# port-channel load-balance src-dst-ip
```

2.16 Lab 41: EtherChannel (PAgP)

Objective

Configure Port Aggregation Protocol (Cisco proprietary).

Configuration Steps

```

1 ! On SW1
2 SW1(config)# interface range gigabitEthernet 0/1-2
3 SW1(config-if-range)# channel-group 1 mode desirable
4 SW1(config-if-range)# exit
5
6 ! On SW2
7 SW2(config)# interface range gigabitEthernet 0/1-2
8 SW2(config-if-range)# channel-group 1 mode desirable
9 SW2(config-if-range)# exit
10
11 ! Verify
12 SW1# show etherchannel summary
13 Group Port-channel Protocol Ports
14 -----+-----+-----+-----
15 1 Po1(SU) PAgP Gi0/1(P) Gi0/2(P)

```

PAgP Modes

Mode	Description
Desirable	Actively negotiates
Auto	Responds passively

Comparison: LACP vs PAgP

Feature	LACP	PAgP
Standard	IEEE 802.3ad	Cisco proprietary
Multi-vendor	Yes	No
Maximum ports	16 (8 active)	8
Mode options	Active/Passive	Desirable/Auto

2.17 Lab 42: Static Routing

Objective

Configure static routes to reach remote networks.

Topology

```
[PC1]---[R1]---(10.0.0.0/30)---[R2]---[PC2]
.1/24      .1/30          .2/30      .1/24
(192.168.1.0/24)                (192.168.2.0/24)
```

Configuration Steps

```
1 ! On R1
2 R1(config)# interface gigabitEthernet 0/0
3 R1(config-if)# ip address 192.168.1.1 255.255.255.0
4 R1(config-if)# no shutdown
5
6 R1(config)# interface serial 0/0/0
7 R1(config-if)# ip address 10.0.0.1 255.255.255.252
8 R1(config-if)# no shutdown
9
10 ! Static route to reach R2's LAN
11 R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
12
13 ! On R2
14 R2(config)# interface gigabitEthernet 0/0
15 R2(config-if)# ip address 192.168.2.1 255.255.255.0
16 R2(config-if)# no shutdown
17
18 R2(config)# interface serial 0/0/0
19 R2(config-if)# ip address 10.0.0.2 255.255.255.252
20 R2(config-if)# no shutdown
21
22 ! Static route back to R1's LAN
23 R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verification

```
1 ! Check routing table
2 R1# show ip route
3 Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
      BGP
4       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
5       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
6       E1 - OSPF external type 1, E2 - OSPF external type 2
7       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
      level-2
8       ia - IS-IS inter area, * - candidate default, U - per-user
      static route
9       o - ODR, P - periodic downloaded static route, H - NHRP, l -
      LISP
10      + - replicated route, % - next hop override
11
```

```
12 Gateway of last resort is not set
13
14 S    192.168.2.0/24 [1/0] via 10.0.0.2
15 C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
16 L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
17 C    10.0.0.0/30 is directly connected, Serial0/0/0
18 L    10.0.0.1/32 is directly connected, Serial0/0/0
19
20 ! Test connectivity
21 PC1> ping 192.168.2.2
22 Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
```

2.18 Lab 43: Default Route Configuration

Objective

Configure default routes for internet-bound traffic.

Topology

[LAN] --- [R1] --- (ISP) --- [Internet]
(.1) (default route)

Configuration Steps

```
1 ! On R1
2 R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
3
4 ! Alternative: Using exit interface
5 R1(config)# ip route 0.0.0.0 0.0.0.0 serial 0/0/0
6
7 ! Using both next-hop and interface
8 R1(config)# ip route 0.0.0.0 0.0.0.0 serial 0/0/0 203.0.113.1
9
10 ! Verify
11 R1# show ip route
12 Gateway of last resort is 203.0.113.1 to network 0.0.0.0
13
14 S*    0.0.0.0/0 [1/0] via 203.0.113.1
15
16 ! Test
17 R1# ping 8.8.8.8
18 !!!!!
```

Default Route Use Cases

- Internet connectivity
- Stub networks (single exit point)
- Simplifying routing tables
- Combined with dynamic routing

2.19 Lab 44: Floating Static Routes

Objective

Configure backup routes with higher administrative distance.

Topology

```
[R1]---(Primary: Serial0/0/0, AD=1)---[R2]
|                                     |
|---(Backup: Serial0/0/1, AD=5)-----|
```

Configuration Steps

```
1 ! Primary route (AD default is 1)
2 R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
3
4 ! Backup route (AD 5 - higher than OSPF's 110)
5 R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.1.2 5
6
7 ! Verify primary route
8 R1# show ip route 192.168.2.0
9 Routing entry for 192.168.2.0/24
10   Known via "static", distance 1, metric 0
11   * 10.0.0.2
12
13 ! Simulate primary link failure
14 R1(config)# interface serial 0/0/0
15 R1(config-if)# shutdown
16
17 ! Check backup route now active
18 R1# show ip route 192.168.2.0
19 Routing entry for 192.168.2.0/24
20   Known via "static", distance 5, metric 0
21   * 10.0.1.2
22
23 ! Restore primary
24 R1(config-if)# no shutdown
25 ! Route reverts to primary
```

Administrative Distance Values

Routing Protocol	AD
Connected interface	0
Static route	1
EIGRP summary	5
External BGP	20
EIGRP internal	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200
Unknown	255

2.20 Lab 45: RIP v2 Configuration

Objective

Configure RIPv2 for small network routing.

Topology

[R1] --- (10.0.0.0/30) --- [R2] --- (10.0.1.0/30) --- [R3]
(192.168.1.0/24) (192.168.3.0/24)
(192.168.2.0/24 on R2 loopback)

Configuration Steps

```
1 ! On R1
2 R1(config)# router rip
3 R1(config-router)# version 2
4 R1(config-router)# no auto-summary
5 R1(config-router)# network 192.168.1.0
6 R1(config-router)# network 10.0.0.0
7 R1(config-router)# exit
8
9 ! On R2
10 R2(config)# router rip
11 R2(config-router)# version 2
12 R2(config-router)# no auto-summary
13 R2(config-router)# network 10.0.0.0
14 R2(config-router)# network 10.0.1.0
15 R2(config-router)# network 192.168.2.0
16 R2(config-router)# exit
17
18 ! On R3
19 R3(config)# router rip
20 R3(config-router)# version 2
21 R3(config-router)# no auto-summary
22 R3(config-router)# network 192.168.3.0
23 R3(config-router)# network 10.0.1.0
```

Verification

```
1 ! Check RIP database
2 R1# show ip rip database
3 192.168.1.0/24      auto-summary
4 192.168.1.0/24      directly connected, GigabitEthernet0/0
5 192.168.2.0/24      auto-summary
6 192.168.2.0/24
7   [1] via 10.0.0.2, 00:00:25, Serial0/0/0
8 192.168.3.0/24      auto-summary
9 192.168.3.0/24
10  [2] via 10.0.0.2, 00:00:25, Serial0/0/0
11
12 ! Check RIP neighbors
13 R1# show ip rip neighbors
14 Neighbor          V      Version      Interface      Address
15 10.0.0.2           2      2              Serial0/0/0    10.0.0.2
```

```
16
17 ! Debug RIP (use cautiously)
18 R1# debug ip rip
19 RIP protocol debugging is on
20 R1#
21 *Mar 15 14:30:15.123: RIP: sending v2 update to 224.0.0.9 via Serial0
    /0/0
22 *Mar 15 14:30:15.123: RIP: build update entries
23 *Mar 15 14:30:15.123:   192.168.1.0/24 via 0.0.0.0, metric 1, tag 0
```

2.21 Lab 46: RIP Troubleshooting

Objective

Troubleshoot common RIP configuration issues.

Common Issues

Issue	Solution
No routes exchanged	Verify no auto-summary
RIPv1 vs RIPv2 mismatch	Ensure both use version 2
Split horizon issues	Check no ip split-horizon on frame relay
Route flapping	Increase timers: timer basic 30 90 120 180
Passive interfaces	Check passive-interface default
Authentication fails	Configure key chain on both sides

Troubleshooting Commands

```

1 ! Verify RIP is running
2 R1# show ip protocols
3 Routing Protocol is "rip"
4   Sending updates every 30 seconds, next due in 18 seconds
5   Invalid after 180 seconds, hold down 180, flushed after 240
6   Outgoing update filter list for all interfaces is not set
7   Incoming update filter list for all interfaces is not set
8   Redistributing: rip
9   Default version control: send version 2, receive version 2
10  Interface          Send Recv  Triggered RIP  Key-chain
11  GigabitEthernet0/0  2      2
12  Serial0/0/0        2      2
13  Automatic network summarization is not in effect
14  Maximum path: 4
15  Routing for Networks:
16    10.0.0.0
17    192.168.1.0
18  Passive Interface(s):
19  Routing Information Sources:
20    Gateway          Distance    Last Update
21    10.0.0.2          120        00:00:03
22  Distance: (default is 120)
23
24 ! Check interface for RIP
25 R1# show ip rip interface
26 Serial0/0/0        (RIP version 2)
27   Sending          (updates every 30 sec)
28   Next due in 12 seconds
29   Invalid after 180 sec
30   Hold down 180 sec
31   Flushed after 240 sec

```


2.22 Lab 47: Single-Area OSPF

Objective

Configure OSPF in a single area (Area 0).

Topology

```
[R1]---(10.0.0.0/30)---[R2]---(10.0.1.0/30)---[R3]
(.1/24)          (.2/.1)          (.2/24)
(Lo0:1.1.1.1/32)  (Lo0:2.2.2.2/32)  (Lo0:3.3.3.3/32)
```

Configuration Steps

```
1 ! On R1
2 R1(config)# router ospf 1
3 R1(config-router)# router-id 1.1.1.1
4 R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
5 R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
6 R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
7 R1(config-router)# exit
8
9 ! On R2
10 R2(config)# router ospf 1
11 R2(config-router)# router-id 2.2.2.2
12 R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
13 R2(config-router)# network 10.0.1.0 0.0.0.3 area 0
14 R2(config-router)# network 2.2.2.2 0.0.0.0 area 0
15 R2(config-router)# exit
16
17 ! On R3
18 R3(config)# router ospf 1
19 R3(config-router)# router-id 3.3.3.3
20 R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
21 R3(config-router)# network 10.0.1.0 0.0.0.3 area 0
22 R3(config-router)# network 3.3.3.3 0.0.0.0 area 0
```

Verification

```
1 ! Check OSPF neighbors
2 R1# show ip ospf neighbor
3
4 Neighbor ID      Pri    State           Dead Time   Address
   Interface
5 2.2.2.2          1     FULL/DR         00:00:35   10.0.0.2
   Serial0/0/0
6
7 ! Check OSPF routes
8 R1# show ip route ospf
9 0       192.168.3.0/24 [110/65] via 10.0.0.2, 00:00:10, Serial0/0/0
10 0       2.2.2.2/32  [110/65] via 10.0.0.2, 00:00:10, Serial0/0/0
11 0       3.3.3.3/32  [110/129] via 10.0.0.2, 00:00:10, Serial0/0/0
12
13 ! Check OSPF database
14 R1# show ip ospf database
```

```
15
16 ! Test connectivity
17 R1# ping 192.168.3.1 source 192.168.1.1
18 !!!!!
```

2.23 Lab 48: OSPF Cost Manipulation

Objective

Change OSPF path selection by modifying interface costs.

Configuration Steps

```
1 ! Method 1: Set cost directly on interface
2 R1(config)# interface serial 0/0/0
3 R1(config-if)# ip ospf cost 100
4
5 ! Method 2: Change reference bandwidth
6 R1(config-router)# auto-cost reference-bandwidth 10000
7 % OSPF: Reference bandwidth is changed.
8     Please ensure reference bandwidth is consistent across all
9     routers.
10
11 ! Method 3: Change interface bandwidth (affects cost)
12 R1(config-if)# bandwidth 1000
13 ! Cost = Reference_BW / Bandwidth (default ref=100Mbps)
14
15 ! Verify costs
16 R1# show ip ospf interface serial 0/0/0
17 Serial0/0/0 is up, line protocol is up
18   Internet Address 10.0.0.1/30, Area 0
19   Process ID 1, Router ID 1.1.1.1, Network Type POINT_TO_POINT, Cost:
20   100
21
22 ! See entire OSPF topology
23 R1# show ip ospf topology
24
25 ! OSPF path selection order:
26 ! 1. Intra-area routes (0)
27 ! 2. Inter-area routes (0 IA)
28 ! 3. E1 external routes
29 ! 4. E2 external routes
```

Reference Bandwidth Best Practices

```
1 ! For modern networks with 10Gbps+
2 R1(config-router)# auto-cost reference-bandwidth 100000
3 ! This sets 10Gbps to cost 1
```

2.24 Lab 49: OSPF Router ID Configuration

Objective

Configure and manage OSPF Router IDs.

Configuration Steps

```
1 ! Method 1: Manual Router ID
2 R1(config)# router ospf 1
3 R1(config-router)# router-id 1.1.1.1
4
5 ! Method 2: Loopback interface (highest IP)
6 R1(config)# interface loopback 0
7 R1(config-if)# ip address 10.255.255.1 255.255.255.255
8
9 ! Method 3: Highest active interface IP (if no loopback)
10
11 ! Router ID selection order:
12 ! 1. Manual router-id command
13 ! 2. Highest loopback IP address
14 ! 3. Highest active interface IP address
15
16 ! Change Router ID (requires OSPF restart)
17 R1# clear ip ospf process
18 Reset ALL OSPF processes? [no]: yes
19
20 ! Verify Router ID
21 R1# show ip ospf
22 Routing Process "ospf 1" with ID 1.1.1.1
23 Start time: 00:00:15.123, Time elapsed: 00:10:00.456
24 Supports only single TOS(TOS0) routes
25 Supports opaque LSA
26 Supports Link-local Signaling (LLS)
27 Supports area transit capability
28
29 ! Check neighbor Router IDs
30 R1# show ip ospf neighbor
31 Neighbor ID      Pri   State           Dead Time   Address
32 2.2.2.2          1    FULL/DR        00:00:35    10.0.0.2
   Serial0/0/0
```

2.25 Lab 50: Passive Interfaces in OSPF

Objective

Configure passive interfaces to stop OSPF hellos on LAN segments.

Configuration Steps

```
1 ! Make LAN interface passive (stops OSPF advertisements but still
   includes network)
2 R1(config)# router ospf 1
3 R1(config-router)# passive-interface gigabitEthernet 0/0
4
5 ! Make all interfaces passive, then enable specific ones
6 R1(config-router)# passive-interface default
7 R1(config-router)# no passive-interface serial 0/0/0
8
9 ! Verify passive interfaces
10 R1# show ip ospf interface
11 GigabitEthernet0/0 is up, line protocol is up
12   Internet Address 192.168.1.1/24, Area 0
13   Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
14   **Passive interface, No hellos**
15
16 Serial0/0/0 is up, line protocol is up
17   Internet Address 10.0.0.1/30, Area 0
18   Process ID 1, Router ID 1.1.1.1, Network Type POINT_TO_POINT, Cost:
19   64
20   Transmit Delay is 1 sec, State POINT_TO_POINT
21   Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
22 ! Check OSPF neighbors (only active interfaces)
23 R1# show ip ospf neighbor
```

Benefits of Passive Interfaces

- Reduces OSPF overhead
- Improves security (no OSPF adjacencies on LAN)
- Still advertises LAN subnet
- Best practice for end-user segments

2.26 Lab 51: OSPF Troubleshooting

Objective

Troubleshoot common OSPF neighbor and route issues.

Common OSPF Issues

Issue	Solution
Neighbors stuck in INIT	Verify multicast reachability (224.0.0.5/6)
Neighbors stuck in 2-WAY	Check DR/BDR election (need Full on broadcast)
Neighbors stuck in EXSTART/EXCHANGE	MTU mismatch, check MTU on interfaces
No routes in table	Verify network statements, area ID matching
Authentication failure	Configure same key/password on both ends
Area mismatch	Check area ID (must match)

Troubleshooting Commands

```
1 ! Check neighbor state
2 R1# show ip ospf neighbor
3 R1# show ip ospf neighbor detail
4
5 ! Check OSPF interface settings
6 R1# show ip ospf interface
7 R1# show ip ospf interface serial 0/0/0
8
9 ! Debug OSPF (use cautiously)
10 R1# debug ip ospf adj
11 R1# debug ip ospf events
12 R1# debug ip ospf packet
13
14 ! Common MTU mismatch debug output
15 *Mar 15 14:30:15.123: OSPF-1 ADJ Ser0/0/0: Send DBD to 10.0.0.2
16 *Mar 15 14:30:15.124: OSPF-1 ADJ Ser0/0/0: Rcv DBD from 10.0.0.2
17 *Mar 15 14:30:15.124: OSPF-1 ADJ Ser0/0/0: Nbr 2.2.2.2 has larger MTU
18 ! Fix with:
19 R1(config-if)# mtu 1500
20
21 ! Clear OSPF process
22 R1# clear ip ospf process
23
24 ! Check OSPF database for inconsistencies
25 R1# show ip ospf database
26 R1# show ip ospf database router
27 R1# show ip ospf database network
```

2.27 Lab 52: EIGRP Basic Configuration

Objective

Configure EIGRP for basic routing.

Configuration Steps

```

1 ! On R1
2 R1(config)# router eigrp 100
3 R1(config-router)# network 192.168.1.0 0.0.0.255
4 R1(config-router)# network 10.0.0.0 0.0.0.3
5 R1(config-router)# network 1.1.1.1 0.0.0.0
6 R1(config-router)# no auto-summary
7 R1(config-router)# exit
8
9 ! On R2
10 R2(config)# router eigrp 100
11 R2(config-router)# network 10.0.0.0 0.0.0.3
12 R2(config-router)# network 10.0.1.0 0.0.0.3
13 R2(config-router)# network 2.2.2.2 0.0.0.0
14 R2(config-router)# no auto-summary
15
16 ! On R3
17 R3(config)# router eigrp 100
18 R3(config-router)# network 192.168.3.0 0.0.0.255
19 R3(config-router)# network 10.0.1.0 0.0.0.3
20 R3(config-router)# network 3.3.3.3 0.0.0.0
21 R3(config-router)# no auto-summary

```

Verification

```

1 ! Check EIGRP neighbors
2 R1# show ip eigrp neighbors
3 EIGRP-IPv4 Neighbors for AS(100)
4 H      Address                Interface           Hold Uptime      SRTT      RTT    Q   Seq
5                               (sec)            (ms)          Cnt  Num
6 0      10.0.0.2                Se0/0/0            11 00:00:15    1      100    0   3
7
8 ! Check EIGRP topology
9 R1# show ip eigrp topology
10 EIGRP-IPv4 Topology Table for AS(100)/ID(1.1.1.1)
11 Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
12         r - reply Status, s - sia Status
13
14 P 192.168.1.0/24, 1 successors, FD is 28160
15     via Connected, GigabitEthernet0/0
16 P 192.168.3.0/24, 1 successors, FD is 2172416
17     via 10.0.0.2 (2172416/2169856), Serial0/0/0
18
19 ! Check EIGRP routes
20 R1# show ip route eigrp
21 D      192.168.3.0/24 [90/2172416] via 10.0.0.2, 00:00:10, Serial0/0/0

```

2.28 Lab 53: EIGRP Metrics and Feasible Distance

Objective

Understand EIGRP metric calculation and successor selection.

Configuration Steps

```
1 ! View EIGRP topology with metrics
2 R1# show ip eigrp topology all-links
3 EIGRP-IPv4 Topology Table for AS(100)/ID(1.1.1.1)
4 Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
5       r - reply Status, s - sia Status
6
7 P 192.168.3.0/24, 1 successors, FD is 2172416, serno 3
8     via 10.0.0.2 (2172416/2169856), Serial0/0/0
9     via 10.0.1.2 (2172416/2169856), Serial0/0/1
10
11 ! Modify EIGRP metric weights
12 R1(config-router)# metric weights 0 1 0 1 0 0
13 ! K1=Bandwidth, K2=Load, K3=Delay, K4=Reliability, K5=MTU
14
15 ! Default formula: Metric = (Bandwidth + Delay) * 256
16 ! Bandwidth = 10^7 / min BW in kbps
17 ! Delay = sum of delays in 10s of microseconds
18
19 ! Change interface bandwidth/delay
20 R1(config)# interface serial 0/0/0
21 R1(config-if)# bandwidth 1544
22 R1(config-if)# delay 20000
23
24 ! Verify metric changes
25 R1# show ip eigrp topology 192.168.3.0/24
26 EIGRP-IPv4 (AS 100): Topology entry for 192.168.3.0/24
27   State is Passive, Query origin flag is 1, 1 Successor(s), FD is
28     40512000
29   Descriptor Blocks:
30     10.0.0.2 (Serial0/0/0), from 10.0.0.2, Send flag is 0x0
31     Composite metric is (40512000/28160), Route is Internal
32     Vector metric:
33       Minimum bandwidth is 1544 Kbit
34       Total delay is 21000 microseconds
35       Reliability is 255/255
36       Load is 1/255
37       Minimum MTU is 1500
38       Hop count is 1
```


2.29 Lab 54: EIGRP Neighbor Troubleshooting

Objective

Troubleshoot EIGRP neighbor relationships.

Common EIGRP Issues

Issue	Solution
AS number mismatch	Verify AS number matches on all routers
K-values mismatch	Check metric weights (must match)
Authentication failure	Configure same key chain
Network statements missing	Verify networks are included
Split horizon blocking	Disable on hub-and-spoke: <code>no ip split-horizon eigrp 100</code>
Passive interface	Check if interface is passive

Troubleshooting Commands

```

1 ! Check neighbor status
2 R1# show ip eigrp neighbors
3 ! If no neighbors, check:
4 ! 1. Interface status
5 R1# show ip interface brief
6
7 ! 2. EIGRP process
8 R1# show ip protocols
9
10 ! 3. EIGRP interface settings
11 R1# show ip eigrp interfaces
12
13 ! Debug EIGRP
14 R1# debug eigrp packets
15 R1# debug eigrp neighbors
16
17 ! K-value mismatch example
18 R1# show ip eigrp neighbors
19 *Mar 15 14:30:15.123: EIGRP: Neighbor 10.0.0.2 not on common network
    for AS 100
20 *Mar 15 14:30:15.124: EIGRP: K-values mismatch
21 ! Fix by matching metric weights
22 R1(config-router)# metric weights 0 1 0 1 0 0
23 R2(config-router)# metric weights 0 1 0 1 0 0
24
25 ! Clear EIGRP neighbors
26 R1# clear ip eigrp neighbors

```

2.30 Lab 55: DHCP Configuration on Router

Objective

Configure router as DHCP server for LAN clients.

Topology

```
[PC1]---[Switch]---[R1]
(DHCP Client)      (DHCP Server)
```

Configuration Steps

```
1 ! Create DHCP pool
2 R1(config)# ip dhcp pool LAN_POOL
3 R1(dhcp-config)# network 192.168.1.0 255.255.255.0
4 R1(dhcp-config)# default-router 192.168.1.1
5 R1(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
6 R1(dhcp-config)# domain-name ccnadomain.com
7 R1(dhcp-config)# lease 7 0 0
8 R1(dhcp-config)# exit
9
10 ! Exclude static addresses
11 R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
12 R1(config)# ip dhcp excluded-address 192.168.1.254
13
14 ! Configure interface for DHCP relay (if needed)
15 R1(config)# interface gigabitEthernet 0/0
16 R1(config-if)# ip address 192.168.1.1 255.255.255.0
17 R1(config-if)# no shutdown
18
19 ! Verify DHCP
20 R1# show ip dhcp binding
21 IP address      Client-ID/      Lease expiration      Type
22                  Hardware address
23 192.168.1.11    0063.6973.636f.0001.  Mar 22 2024 02:30 PM
24                  Automatic
25                  000a.000b.000c--.ab12
26
27 R1# show ip dhcp pool
28 Pool LAN_POOL :
29 Utilization mark (high/low)      : 100 / 0
30 Subnet size (first/next)          : 0 / 0
31 Total addresses                   : 254
32 Leased addresses                  : 1
33 Excluded addresses                 : 11
34 Pending event                     : none
35 1 subnet is currently in the pool
36 Current index      IP address range      Leased
37 addresses
38 192.168.1.11      192.168.1.1      - 192.168.1.254    1
39
40 ! Test on PC
41 PC> ipconfig /dhcp
42 IP Address: 192.168.1.11
43 Subnet Mask: 255.255.255.0
44 Default Gateway: 192.168.1.1
```

43 **DNS Servers: 8.8.8.8, 8.8.4.4**

2.31 Lab 56: DHCP Relay Agent

Objective

Configure DHCP relay to forward requests across subnets.

Topology

[PC1] --- [R1] --- (Network) --- [R2] --- [DHCP Server]
(VLAN10) (192.168.100.10)

Configuration Steps

```
1 ! On R1 (relay agent)
2 R1(config)# interface gigabitEthernet 0/0
3 R1(config-if)# ip address 192.168.1.1 255.255.255.0
4 R1(config-if)# ip helper-address 192.168.100.10
5 R1(config-if)# exit
6
7 ! On R2 (router between)
8 R2(config)# interface gigabitEthernet 0/0
9 R2(config-if)# ip address 192.168.100.1 255.255.255.0
10 R2(config-if)# no shutdown
11
12 ! On DHCP Server (can be router or actual server)
13 ! Configure DHCP pool for remote subnet
14 DHCP-Server(config)# ip dhcp pool REMOTE_POOL
15 DHCP-Server(dhcp-config)# network 192.168.1.0 255.255.255.0
16 DHCP-Server(dhcp-config)# default-router 192.168.1.1
17 DHCP-Server(dhcp-config)# exit
18
19 ! Verify relay
20 R1# show ip interface gigabitEthernet 0/0 | include helper
21 Helper address is 192.168.100.10
22
23 ! Debug DHCP relay
24 R1# debug ip dhcp server events
25 R1# debug ip udp 67
```

Ports Used by DHCP

Port	Usage
UDP 67	DHCP server
UDP 68	DHCP client

2.32 Lab 57: DNS Configuration Basics

Objective

Configure DNS on router and test name resolution.

Configuration Steps

```
1 ! Configure DNS server on router
2 R1(config)# ip domain-lookup
3 R1(config)# ip name-server 8.8.8.8
4 R1(config)# ip name-server 8.8.4.4
5
6 ! Configure static DNS entry
7 R1(config)# ip host R2 10.0.0.2
8 R1(config)# ip host webserver 192.168.1.100
9 R1(config)# ip host mailserver 192.168.1.101
10
11 ! Verify DNS resolution
12 R1# ping R2
13 Translating "R2"...domain server (8.8.8.8) [OK]
14 Type escape sequence to abort.
15 Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
16 !!!!!
17
18 ! Test DNS lookup
19 R1# ping webserver
20 Translating "webserver"...domain server (8.8.8.8) [OK]
21 Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:
22 .....
23
24 ! Show DNS cache
25 R1# show hosts
26 Default domain is not set
27 Name/address lookup uses domain service
28 Name servers are 8.8.8.8, 8.8.4.4
29
30 Host                Flags          Age Type      Address(es)
31 R2                  (temp) (0) 0      IP        10.0.0.2
32 webserver          (perm) (0) 0      IP        192.168.1.100
33 mailserver         (perm) (0) 0      IP        192.168.1.101
34
35 ! Clear DNS cache
36 R1# clear host
```

2.33 Lab 58: NTP Configuration

Objective

Configure NTP for accurate time synchronization.

Topology

```
[R1]---(Network)---[NTP Server]
(NTP Client)         (192.168.1.100)
```

Configuration Steps

```
1 ! Configure NTP client
2 R1(config)# ntp server 192.168.1.100
3 R1(config)# ntp server 0.pool.ntp.org
4
5 ! Configure router as NTP server
6 R1(config)# ntp master 2
7 ! (stratum level, lower is more accurate)
8
9 ! Authenticate NTP
10 R1(config)# ntp authenticate
11 R1(config)# ntp authentication-key 1 md5 cisco123
12 R1(config)# ntp trusted-key 1
13 R1(config)# ntp server 192.168.1.100 key 1
14
15 ! Verify NTP
16 R1# show ntp status
17 Clock is synchronized, stratum 3, reference is 192.168.1.100
18 nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is
19 2**10
20 reference time is DF1F7B5C.12345678 (14:30:15.123 EST Thu Mar 15 2024)
21 clock offset is 0.1234 msec, root delay is 12.34 msec
22 root dispersion is 23.45 msec, peer dispersion is 5.67 msec
23 loopfilter state is 'CTRL' (Normal Controlled Loop), drift is
24 0.0000123456 s/s
25
26 ! Show NTP associations
27 R1# show ntp associations
28
29 address          ref clock      st    when    poll    reach    delay
30 offset    disp
31 *~192.168.1.100  .GPS.          1     23     64     377     12.34
32 0.12      5.67
33 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~
34 configured
```

2.34 Lab 59: Syslog Configuration

Objective

Configure syslog to send logs to central server.

Configuration Steps

```
1 ! Configure syslog server
2 R1(config)# logging host 192.168.1.100
3 R1(config)# logging trap informational
4 R1(config)# logging source-interface loopback 0
5
6 ! Logging levels (0-7)
7 ! 0 - Emergency (system unusable)
8 ! 1 - Alert (immediate action)
9 ! 2 - Critical (critical condition)
10 ! 3 - Error (error condition)
11 ! 4 - Warning (warning condition)
12 ! 5 - Notification (normal but significant)
13 ! 6 - Informational (informational messages)
14 ! 7 - Debugging (debug messages)
15
16 ! Configure logging options
17 R1(config)# logging on
18 R1(config)# service timestamps log datetime msec localtime
19 R1(config)# logging buffered 16384 informational
20 R1(config)# logging console warnings
21 R1(config)# logging monitor debugging
22
23 ! Verify logging
24 R1# show logging
25 Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0
   flushes, 0 overruns)
26   Console logging: level warnings (4)
27   Monitor logging: level debugging (7)
28   Buffer logging: level informational (6), 16384 bytes logged
29   Trap logging: level informational (6), 32 message lines logged
30     Logging to 192.168.1.100 (udp port 514, audit disabled,
31       authentication disabled, encryption disabled)
32
33 ! Generate test log
34 R1# send log "Test syslog message"
```

2.35 Lab 60: SNMP Basics

Objective

Configure SNMP for network monitoring.

Configuration Steps

```
1 ! Configure SNMP community strings
2 R1(config)# snmp-server community public R0
3 R1(config)# snmp-server community private RW
4
5 ! Set SNMP location and contact
6 R1(config)# snmp-server location "Data Center - Rack A"
7 R1(config)# snmp-server contact "admin@ccnadomain.com"
8
9 ! Configure SNMP traps
10 R1(config)# snmp-server enable traps
11 R1(config)# snmp-server host 192.168.1.100 version 2c public
12
13 ! SNMP versions
14 ! v1 - Original, least secure
15 ! v2c - Uses community strings (public/private)
16 ! v3 - Authentication and encryption
17
18 ! SNMPv3 configuration
19 R1(config)# snmp-server group ADMIN v3 priv
20 R1(config)# snmp-server user admin ADMIN v3 auth md5 cisco123 priv des
    cisco123
21
22 ! Verify SNMP
23 R1# show snmp community
24 Community name: public
25 Community name: private
26
27 R1# show snmp group
28 groupname: ADMIN                                security model:v3 priv
29 contextname: <no context specified>              storage-type: permanent
30 readview : <no readview specified>                writeview: <no writeview
    specified>
31 notifyview: <no notifyview specified>
32 row status: active
33
34 R1# show snmp user
35 User name: admin
36 Engine ID: 800000090300001122334455
37 storage-type: permanent
38 Authentication Protocol: MD5
39 Privacy Protocol: DES
40 Group-name: ADMIN
```


Appendix

A: Cisco IOS Command Hierarchy

```
User EXEC Mode (>)           - Limited monitoring
    | enable
Privileged EXEC Mode (#)      - Full access to device
    | configure terminal
Global Configuration Mode     - Device-wide settings
    | interface <name>
Interface Configuration Mode - Port-specific settings
    | router <protocol>
Routing Protocol Mode         - Protocol-specific settings
    | line <type> <range>
Line Configuration Mode       - Console/VTY settings
```

B: Common Port Numbers

Port	Service
TCP/20,21	FTP
TCP/22	SSH
TCP/23	Telnet
TCP/25	SMTP
UDP/53	DNS
UDP/67,68	DHCP
UDP/69	TFTP
TCP/80	HTTP
TCP/110	POP3
UDP/123	NTP
TCP/443	HTTPS
UDP/514	Syslog
UDP/161,162	SNMP

C: Subnetting Cheat Sheet

Prefix	Subnet Mask	Wildcard Mask
/8	255.0.0.0	0.255.255.255
/16	255.255.0.0	0.0.255.255
/24	255.255.255.0	0.0.0.255
/25	255.255.255.128	0.0.0.127
/26	255.255.255.192	0.0.0.63
/27	255.255.255.224	0.0.0.31
/28	255.255.255.240	0.0.0.15
/29	255.255.255.248	0.0.0.7
/30	255.255.255.252	0.0.0.3
/31	255.255.255.254	0.0.0.1
/32	255.255.255.255	0.0.0.0

D: Useful Shortcuts

```
1 Ctrl+Shift+6 (then X) - Suspend Telnet/SSH session
2 Ctrl+Z - Exit config mode to privileged EXEC
3 Ctrl+C - Abort command
4 Ctrl+R - Redisplay line
5 Tab - Complete command
6 ? - Context-sensitive help
```

E: Lab Completion Checklist

Use this checklist to track your progress through all 100 labs:

Level	Labs Completed
Beginner (1-25)	___/25
Intermediate - Switching (26-41)	___/16
Intermediate - Routing (42-54)	___/13
Intermediate - Services (55-60)	___/6
Advanced - NAT & ACL (61-69)	___/9
Advanced - WAN & Routing (70-79)	___/10
Advanced - Security (80-86)	___/7
Advanced - Services (87-91)	___/5
Advanced - Troubleshooting (92-100)	___/9

F: Recommended Resources

- Cisco Packet Tracer (Free for NetAcad students)
- GNS3 (Advanced, requires IOS images)
- EVE-NG (Professional virtual lab)
- Cisco Modeling Labs (CML) - Paid
- Boson NetSim (Exam simulation)

Conclusion

Congratulations on completing all 100 CCNA labs! You have now mastered:

- Basic device configuration and management
- Switching technologies (VLANs, STP, EtherChannel)
- Routing protocols (Static, RIP, OSPF, EIGRP)
- Network services (DHCP, DNS, NTP, Syslog)
- Security features (SSH, ACLs, Port Security)
- WAN technologies (PPP, GRE)
- Troubleshooting methodologies

Next Steps:

1. Review weak areas using your lab notes
2. Take practice exams (Boson, MeasureUp)
3. Build a full enterprise lab combining all technologies
4. Schedule your CCNA 200-301 exam

Good luck on your CCNA journey!